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Assignment 01

I'm here to present my module assignment as I 'm suppose to per Assignment Instructions.

Module code, SME3701-Solid mechanics IV. Most important I will be using **MATLAB**.

Starting with **Question 1.1**

In The question Im asked to

Determine the natural angular frequency ωn of the system using MATLAB.

Before showing the MATLAB, I would like to show you the formula used to determine

The natural angular frequency ωn.

Which is

$$\omega_n = \sqrt{\frac{k}{m}}$$

Given:

- $k = 2000 \text{ N/m}$
- $m = 5 \text{ kg}$

K =stands for the spring with stiffness which is 2000 N/m.

While m =mass =5 kg.

Below is MATLAB CODE.

```
m = 5;
k = 2000;

wn = sqrt(k/m)
```

I will press F5 or enter, it will give me the result of the natural angular frequency ωn = 20

So, the natural angular frequency is:

$$\omega_n = 20 \text{ rad/s}$$

Now I am going to do **Question 1.2 im required to**

1.2 Develop a MATLAB script to compute the displacement (t) of the system.

Bu before I start to solve this question, I would like to show you the formula used to compute the displacement $x(t)$ of the system. Below as follows

As given the system is initially displacement is Xo= 0.02 m.

And we had already calculated and obtained

So, the natural angular frequency is:

$$\omega_n = 20 \text{ rad/s}$$

So the formula used

For an undamped system with zero initial velocity, the displacement is:

$$x(t) = x_0 \cos(\omega_n t)$$

Given:

- $x_0 = 0.02 \text{ m}$
- $\omega_n = 20 \text{ rad/s}$

Below is the MAITLAB script and This script defines the system parameters, computes the natural frequency, creates a time vector, and calculates the displacement x(t) over time:

```
% Parameters
m = 5;
k = 2000;
wn = sqrt(k/m);

x0 = 0.02; % initial displacement (m)
t = Lin space(0,5,1000); % time vector (0 to 5 s)

% Displacement
x = x0*cos(wn*t);
```

Above script defines the system parameters, computes the natural frequency, creates a time vector, and calculates the displacement x(t) over time.

Question 1.3

Below is question 1.3 and *im required to Plot the displacement versus time for the interval 0 ≤ t ≤ 5 seconds using MATLAB.*

I will show you all steps and will plot displayed.

To plot displacement versus time over 0 ≤ t ≤5 seconds, you simply use the displacement you computed and generate a graph.

Below is MATLAB Script:

```
% Parameters
m = 5;
k = 2000;
wn = sqrt(k/m);

x0 = 0.02;
t = linspace(0,5,1000);

% Displacement
x = x0*cos(wn*t);

% Plot
plot(t, x, 'b', 'LineWidth', 1.5);
xlabel('Time (s)');
ylabel('Displacement x(t) (m)');
title('Displacement vs Time');
grid on;
```

As you can see graph is displayed using above MATLAB Script.

What the plot shows:

- A **cosine wave** oscillating between 0.02 m and −0.02 m
- Constant amplitude (no decay, since damping = 0)
- Frequency determined by ω_n=20 rad/s

This confirms the system undergoes steady **simple harmonic motion**.

1.4

Below is question 1.4

The velocity is the time derivative of displacement. For $x(t) = x_0 \cos(\omega_n t)$, the velocity is:

$$v(t) = -x_0 \, \omega_n \sin(\omega_n t)$$

Given:

- $x_0 = 0.02 \text{ m}$
- $\omega_n = 20 \text{ rad/s}$

Here shows MATLAB Script (velocity + separate plot):

```
% Parameters
m = 5;
k = 2000;
wn = sqrt(k/m);

x0 = 0.02;
t = linspace(0,5,1000);

% Velocity
v = -x0*wn*sin(wn*t);

% Plot velocity
figure;
plot(t, v, 'r', 'LineWidth', 1.5);
xlabel('Time (s)');
ylabel('Velocity v(t) (m/s)');
title('Velocity vs Time');
grid on;
```

What the plot shows:

- A **sine wave** (shifted relative to displacement)
- Maximum velocity occurs when displacement is zero
- Velocity is **90° out of phase** with displacement

This confirms the expected behaviour of an undamped simple harmonic oscillator.

Here is Question 1.5

Nature of the motion

The system undergoes simple harmonic motion (SHM). The displacement varies sinusoidally with time, meaning the mass oscillates back and forth about the equilibrium position at a constant frequency and amplitude.

Effect of zero damping

Since the system is undamped:

- ✓ There is no energy loss (no friction or resistance).
- ✓ The total mechanical energy remains constant.
- ✓ The oscillations continue indefinitely with constant amplitude.
- ✓ The motion is perfectly periodic.

Relationship between displacement and velocity

- Velocity is the time derivative of displacement.
- Displacement follows a cosine function, while velocity follows a sine function.
- Therefore, velocity is **90° ($\pi/2$ radians)** out of phase with displacement:

    - ✓ When displacement is maximum, velocity is zero.
    - ✓ When displacement is zero, velocity is maximum.

- Velocity changes sign depending on the direction of motion.

These characteristics are typical of an ideal undamped mass–spring system.